

Problem Set 1 — Solutions

Labor Economics (Saggio), Summer 2026

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Question 1: Returns to Schooling

An infinitely-lived agent picks schooling length S , faces discount rate r , earns the scholarship flow T on $[0, S]$, and earns $y(S)$ forever after date S . She maximises the present value of earnings plus transfers.

Part 1: Objective and optimality condition

In continuous time the objective is

$$V(S) = \underbrace{\int_0^S T e^{-rt} dt}_{\text{scholarship while in school}} + \underbrace{\int_S^\infty y(S) e^{-rt} dt}_{\text{earnings after school}} = \frac{T(1 - e^{-rS})}{r} + \frac{y(S)e^{-rS}}{r}. \quad (1)$$

Differentiating with respect to S ,

$$V'(S) = T e^{-rS} + \frac{y'(S)e^{-rS} - r y(S)e^{-rS}}{r} = e^{-rS} \left[T + \frac{y'(S)}{r} - y(S) \right]. \quad (2)$$

Setting $V'(S) = 0$ and cancelling $e^{-rS} > 0$ gives the optimality condition

$$\boxed{y'(S^*) = r(y(S^*) - T)}. \quad (3)$$

Interpretation. Rewrite as $\frac{y'(S^*)}{r} = y(S^*) - T$. The left-hand side is the *marginal benefit* of one more year of school: the permanent earnings increment $y'(S)$ capitalised at rate r . The right-hand side is the *marginal cost*: the earnings $y(S)$ forgone by staying in school one more instant, *net* of the scholarship T that is still collected. The agent keeps studying until the discounted value of the higher earnings path just balances the net opportunity cost of delaying entry into the labor market. A more generous scholarship T lowers the marginal cost and raises optimal schooling.

Part 2: Ability and the optimal duration of schooling

With $y_i(S) = \exp(\alpha_i + \beta S)$ we have $y'_i(S) = \beta y_i(S)$. The optimality condition becomes

$$\beta y_i = r(y_i - T) \implies y_i(S_i^*) = \frac{rT}{r - \beta}, \quad (4)$$

which requires $r > \beta$ (so that the cost of delay eventually dominates and an interior optimum exists). Taking logs of $\exp(\alpha_i + \beta S_i^*) = rT/(r - \beta)$,

$$S_i^* = \frac{1}{\beta} \left[\log\left(\frac{rT}{r - \beta}\right) - \alpha_i \right]. \quad (5)$$

Do higher-ability agents get more or less schooling? *Less*: the coefficient on α_i is $-1/\beta < 0$. Intuitively, the marginal return to schooling β is common to everyone, but the optimal stopping rule pins the *level* of earnings at $y^* = rT/(r - \beta)$, identical across agents. A high-ability agent starts from a higher α_i , so she reaches that target earnings level with fewer years of school. Equivalently, ability and schooling are substitutes in producing y , and the opportunity cost of an extra year (forgone earnings $y_i(S) - T$) rises faster for the high-ability agent, pushing her out of school sooner.

Part 3: OLS bias with heterogeneous ability and scholarships

Now α_i and T_i vary across agents while β is common. From Part 2, at the optimum $y_i = rT_i/(r - \beta)$, so

$$\log y_i = \log\left(\frac{r}{r - \beta}\right) + \log T_i, \quad (6)$$

$$S_i = \frac{1}{\beta} \left[\log\left(\frac{r}{r - \beta}\right) + \log T_i - \alpha_i \right]. \quad (7)$$

Note $\log y_i$ depends on $\log T_i$ only (not on α_i directly): ability heterogeneity is “undone” in equilibrium earnings because high- α agents choose less schooling. The OLS slope of $\log y$ on S has probability limit $\text{plim } \hat{b} = \text{Cov}(\log y_i, S_i) / \text{Var}(S_i)$. Using $\sigma_T^2 = \text{Var}(\log T_i)$, $\sigma_\alpha^2 = \text{Var}(\alpha_i)$, $\sigma_{T\alpha} = \text{Cov}(\log T_i, \alpha_i)$:

$$\text{Cov}(\log y_i, S_i) = \text{Cov}\left(\log T_i, \frac{1}{\beta}(\log T_i - \alpha_i)\right) = \frac{1}{\beta}(\sigma_T^2 - \sigma_{T\alpha}), \quad (8)$$

$$\text{Var}(S_i) = \frac{1}{\beta^2} \text{Var}(\log T_i - \alpha_i) = \frac{1}{\beta^2}(\sigma_T^2 + \sigma_\alpha^2 - 2\sigma_{T\alpha}). \quad (9)$$

Therefore

$$\text{plim } \hat{b} = \beta \frac{\sigma_T^2 - \sigma_{T\alpha}}{\sigma_T^2 + \sigma_\alpha^2 - 2\sigma_{T\alpha}}. \quad (10)$$

Discussion of the bias.

- *No ability heterogeneity* ($\sigma_\alpha^2 = 0$, $\sigma_{T\alpha} = 0$): $\text{plim } \hat{b} = \beta$. OLS is consistent — all schooling variation is driven by the “cost shifter” T_i , which is independent of the outcome equation, so T_i acts like an instrument and there is no bias.
- *Ability variance* ($\sigma_\alpha^2 > 0$, $\sigma_{T\alpha} = 0$): $\text{plim } \hat{b} = \beta \sigma_T^2 / (\sigma_T^2 + \sigma_\alpha^2) < \beta$. Variation in α_i adds noise to S_i that is unrelated to $\log y_i$, *attenuating* the slope toward zero.
- *Sorting* ($\sigma_{T\alpha} \neq 0$): the covariance term shifts the bias. Because here $\log y_i$ rises with T_i but high-ability agents take *less* schooling, a positive $\sigma_{T\alpha}$ reduces the numerator and denominator in offsetting ways; the net effect depends on the relative magnitudes. The key message is that OLS recovers β only when schooling choices are uncorrelated with ability.

Part 4: OLS bias with heterogeneous returns

Now $T_i = T$ is constant but returns are heterogeneous, $y_i(S) = \exp(\alpha_i + \beta_i S)$. The Part-1 condition $\beta_i y_i = r(y_i - T)$ gives $y_i(S_i^*) = rT/(r - \beta_i)$, so

$$\log y_i = \log(rT) - \log(r - \beta_i), \quad (11)$$

$$S_i = \frac{1}{\beta_i} [\log(rT) - \log(r - \beta_i) - \alpha_i]. \quad (12)$$

This is a *correlated random coefficients* model: the slope β_i that generates earnings is the same object that enters the schooling choice. Because (12) is nonlinear in β_i , we obtain a closed form by a first-order expansion around the means $\beta \equiv E[\beta_i]$, $\alpha \equiv E[\alpha_i]$. Write $\beta_i = \beta + b_i$, $\alpha_i = \alpha + a_i$, with $\text{Var}(b_i) = \sigma_\beta^2$, $\text{Var}(a_i) = \sigma_\alpha^2$, $\text{Cov}(b_i, a_i) = \sigma_{\beta\alpha}$, and define

$$\varphi \equiv \frac{1}{r - \beta}, \quad \bar{S} \equiv \frac{\log(rT) - \log(r - \beta) - \alpha}{\beta}, \quad \psi \equiv \varphi - \bar{S}.$$

Since $\partial[-\log(r - \beta_i)]/\partial\beta_i = \varphi$ at $\beta_i = \beta$, equation (11) linearises to $\log y_i \approx \text{const} + \varphi b_i$, and (12) to

$$S_i - \bar{S} \approx \frac{1}{\beta} [(\varphi - \bar{S}) b_i - a_i] = \frac{1}{\beta} [\psi b_i - a_i].$$

Hence

$$\text{Cov}(\log y_i, S_i) \approx \frac{\varphi}{\beta} (\psi \sigma_\beta^2 - \sigma_{\beta\alpha}), \quad (13)$$

$$\text{Var}(S_i) \approx \frac{1}{\beta^2} (\psi^2 \sigma_\beta^2 + \sigma_\alpha^2 - 2\psi \sigma_{\beta\alpha}), \quad (14)$$

and the OLS slope satisfies

$$\text{plim } \hat{b} \approx \varphi \beta \frac{\psi \sigma_\beta^2 - \sigma_{\beta\alpha}}{\psi^2 \sigma_\beta^2 + \sigma_\alpha^2 - 2\psi \sigma_{\beta\alpha}}, \quad \varphi = \frac{1}{r - \beta}, \quad \psi = \varphi - \bar{S}. \quad (15)$$

(The constants φ and ψ fix the linearisation point; the *bias* is governed by the second moments $\sigma_\beta^2, \sigma_\alpha^2, \sigma_{\beta\alpha}$.)

Discussion of the bias.

- *Homogeneous returns* ($\sigma_\beta^2 = 0$, hence $\sigma_{\beta\alpha} = 0$): $\text{plim } \hat{b} \theta_0$. With T constant and a common β , equation (11) makes $\log y_i$ *identical* across agents (y is pinned to $rT/(r - \beta)$), so there is no earnings variation to explain and the slope collapses to zero. *All* earnings dispersion in this model comes from heterogeneous returns β_i .
- *Return heterogeneity alone* ($\sigma_\beta^2 > 0$, $\sigma_\alpha^2 = \sigma_{\beta\alpha} = 0$): $\text{plim } \hat{b} \approx \varphi \beta / \psi \neq \beta$. Even with no ability heterogeneity, OLS does *not* recover the average return β , because agents with different β_i optimally choose different S_i — selection on gains contaminates the slope.
- *Sorting on gains* ($\sigma_{\beta\alpha} \neq 0$) and ability variance ($\sigma_\alpha^2 > 0$) move the estimand further from β and, as the sign of ψ can be negative, $\text{plim } \hat{b}$ may even take the *opposite sign* of the average return. The practical lesson is the same as Part 3: a bivariate OLS of log earnings on schooling generally fails to identify the causal return when schooling is chosen, and the failure is more severe when the return itself is heterogeneous and correlated with the choice.

Question 2: Keane and Wolpin (1997)

Ages $a \in \{1, 2\}$; choices $m \in \{1, 2, 3\} = \{\text{white collar, blue collar, school}\}$. Flow utilities

$$R_m(a) = \alpha_m + \beta_m G(a) + \gamma_m X_m(a) + \epsilon_m(a) \quad (m = 1, 2), \quad R_3(a) = \epsilon_3(a),$$

with the school alternative normalised ($\alpha_3 = \beta_3 = \gamma_3 = 0$). The $\epsilon_m(a)$ are i.i.d. Type-I Extreme Value across alternatives and time; the period-2 shocks are unknown in period 1. The agent enters period 1 with $G(1) = X_1(1) = X_2(1) = 0$ and discounts the future at $\delta \in (0, 1)$.

Part 1: State vector and period-2 choice probabilities

The state entering age 2 (before $\epsilon_m(2)$ is realised) is

$$S(2) = (G(2), X_1(2), X_2(2)).$$

Because the agent begins period 1 empty and makes a single choice, exactly one state component equals 1 at age 2. Thus $S(2)$ takes only *three* values:

$$\underbrace{(0, 1, 0)}_{\text{worked WC in 1}}, \quad \underbrace{(0, 0, 1)}_{\text{worked BC in 1}}, \quad \underbrace{(1, 0, 0)}_{\text{schooled in 1}}.$$

Period 2 is the last period, so it is a static multinomial-logit choice. Writing the deterministic parts

$$v_1(2) = \alpha_1 + \beta_1 G(2) + \gamma_1 X_1(2), \quad v_2(2) = \alpha_2 + \beta_2 G(2) + \gamma_2 X_2(2), \quad v_3(2) = 0,$$

the Type-I EV assumption gives, for each $S(2)$,

$$\Pr(C(2) = m | S(2)) = \frac{\exp(v_m(2))}{\exp(v_1(2)) + \exp(v_2(2)) + \exp(v_3(2))}, \quad m = 1, 2, 3. \quad (16)$$

Part 2: Expected value function at age 2

With $V_2(S(2)) = \max_m R_m(2; S(2))$ and the Type-I EV hint $E[\max_k U_{ik} + \eta_{ik}] = \nu + \log \sum_k \exp(U_{ik})$, the expectation (prior to the realisation of $\epsilon_m(2)$, dropping the constant ν) is

$$E[V_2(S(2))] = \log(\exp(v_1(2)) + \exp(v_2(2)) + 1), \quad (17)$$

evaluated at each of the three states (note $\exp(v_3(2)) = e^0 = 1$). Explicitly,

$$\begin{aligned} \text{WC experience } (0, 1, 0) : & \quad \log(e^{\alpha_1 + \gamma_1} + e^{\alpha_2} + 1), \\ \text{BC experience } (0, 0, 1) : & \quad \log(e^{\alpha_1} + e^{\alpha_2 + \gamma_2} + 1), \\ \text{Education } (1, 0, 0) : & \quad \log(e^{\alpha_1 + \beta_1} + e^{\alpha_2 + \beta_2} + 1). \end{aligned}$$

Part 3: Period-1 expected payoffs and choice probabilities

In period 1 the deterministic flows are the intercepts only (state is empty): $v_1(1) = \alpha_1$, $v_2(1) = \alpha_2$, $v_3(1) = 0$. Each choice deterministically sets next period's state, so the choice-specific value

(excluding $\epsilon_m(1)$) is the current flow plus the discounted expected continuation value of the state it leads to:

$$w_1 = \alpha_1 + \delta \mathbb{E}[V_2 | (0, 1, 0)] \quad (\text{white collar}), \quad (18)$$

$$w_2 = \alpha_2 + \delta \mathbb{E}[V_2 | (0, 0, 1)] \quad (\text{blue collar}), \quad (19)$$

$$w_3 = 0 + \delta \mathbb{E}[V_2 | (1, 0, 0)] \quad (\text{school}). \quad (20)$$

Adding the i.i.d. Type-I EV shocks $\epsilon_m(1)$, the *ex ante* period-1 choice probabilities are again multinomial logit:

$$\Pr(C(1) = m) = \frac{\exp(w_m)}{\exp(w_1) + \exp(w_2) + \exp(w_3)}, \quad m = 1, 2, 3. \quad (21)$$

School has no current payoff but yields the largest education continuation value $\mathbb{E}[V_2 | (1, 0, 0)]$; the agent trades current earnings against the discounted option value of being more educated (or more experienced) at age 2.

Part 4: Likelihood

Individual i is observed choosing the sequence $(C_i(1), C_i(2))$. The period-1 choice determines $S_i(2)$, so by the law of total probability the individual likelihood factors as

$$L_i(\theta) = \Pr(C(1) = C_i(1)) \cdot \Pr(C(2) = C_i(2) | S_i(2)), \quad (22)$$

where $S_i(2)$ is the state implied by $C_i(1)$. Taking logs and summing over the sample, the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^N \left[\log \Pr(C(1) = C_i(1)) + \log \Pr(C(2) = C_i(2) | S_i(2)) \right], \quad (23)$$

with $\theta = (\alpha_1, \beta_1, \gamma_1, \alpha_2, \beta_2, \gamma_2, \delta)'$ and the probabilities given in Parts 1 and 3.

Part 5: Maximum-likelihood estimation

I estimate θ by maximum likelihood on the $N = 5,000$ simulated choice sequences in `kw_example.csv`, minimising $-\ell(\theta)$ in Python (`scipy.optimize.minimize`, BFGS — the analogue of Matlab's `fminunc`). I search over $\log \delta$ to keep $\delta > 0$, and use several random starting values, retaining the global optimum. The code is in `kw_estimate.py`; standard errors are described below.

Standard errors (extra credit). The MLE $\hat{\theta} \stackrel{a}{\sim} \mathcal{N}(\theta_0, V/N)$, and I report three consistent estimators of the asymptotic variance, all computed on the $\rho = \log \delta$ parameterisation and then mapped to δ by the delta method (see below). Let $s_i(\theta) = \partial \ell_i / \partial \theta$ be the score of observation i and $\hat{H} = -\sum_i \partial^2 \ell_i / \partial \theta \partial \theta'$ the observed (negative) Hessian, both evaluated at $\hat{\theta}$:

$$\underbrace{\hat{V}_H = \hat{H}^{-1}}_{\text{observed information}}, \quad \underbrace{\hat{V}_J = \left(\sum_i s_i s_i' \right)^{-1}}_{\text{OPG / BHHH}}, \quad \underbrace{\hat{V}_S = \hat{H}^{-1} \left(\sum_i s_i s_i' \right) \hat{H}^{-1}}_{\text{sandwich / robust}}. \quad (24)$$

I compute $\hat{H}$ by a finite-difference Hessian of $-\ell$ and the scores s_i by central differences; because each observation's likelihood depends only on its $(C(1), C(2))$ cell, the score takes only nine distinct

Table 1: Maximum-likelihood estimates of the simplified Keane–Wolpin model

Parameter	Estimate	Standard error		
		Hessian	OPG	Sandwich
α_1 (white-collar intercept)	0.4234	0.0585	0.0624	0.0552
β_1 (return to education, WC)	0.3681	0.0800	0.0873	0.0738
γ_1 (return to WC experience)	0.0607	0.0844	0.0881	0.0814
α_2 (blue-collar intercept)	0.4237	0.0422	0.0431	0.0416
β_2 (return to education, BC)	0.1215	0.0635	0.0636	0.0634
γ_2 (return to BC experience)	0.3513	0.0792	0.0823	0.0768
δ (discount factor)	0.8395	0.3818	0.4147	0.3541
Log-likelihood = $-10,734.24$, $N = 5,000$				

values, so the OPG matrix is $\sum_{\text{cells}} n_{\text{cell}} s s'$. The reported baseline SEs are the Hessian-based $\sqrt{\text{diag } \hat{V}_H}$.

Delta method for δ . I optimise over $\rho = \log \delta$; since $\delta = e^\rho$ has $\partial \delta / \partial \rho = \delta$, the SE of δ is $\widehat{\text{SE}}(\delta) = \delta \cdot \widehat{\text{SE}}(\rho)$ (with $\widehat{\text{SE}}(\log \delta) = 0.4548$ under the Hessian estimator).

Information-matrix equality. Under correct specification the information-matrix equality $H = J$ holds, so the three estimators should coincide asymptotically. They do: across the six slope/intercept parameters the Hessian, OPG, and sandwich SEs agree to within about ten percent of one another (differing only in the second decimal), which is reassuring evidence that the model and likelihood are correctly specified. The lone caveat is δ , whose SE (≈ 0.38 – 0.41) is large under every method because the likelihood is flat in δ .

Goodness of fit. With seven parameters for the eight free cell probabilities of the 3×3 table of $(C(1), C(2))$ sequences, the model reproduces every empirical cell frequency to within 0.0014 in probability, confirming the likelihood is correctly specified and the optimum is the global one.

Economic interpretation.

- *Intercepts.* $\alpha_1 \approx \alpha_2 \approx 0.42$: for an inexperienced, uneducated entrant, the two occupations offer almost identical baseline utility, so the period-1 choice is driven largely by their differing *continuation* values.
- *Returns to education.* $\beta_1 = 0.37 \gg \beta_2 = 0.12$: a year of schooling raises white-collar utility about three times as much as blue-collar utility. Education is the human-capital investment that pays off mainly in the white-collar track (β_1 is precisely estimated; β_2 is only marginally distinguishable from zero).
- *Returns to experience.* The pattern reverses for on-the-job experience: $\gamma_2 = 0.35$ (precise) versus $\gamma_1 = 0.06$ (indistinguishable from zero). Blue-collar work builds valuable occupation-specific human capital, whereas white-collar experience adds little.
- *Discount factor.* $\hat{\delta} = 0.84 \in (0, 1)$ is economically reasonable, but its standard error is large (0.38): the log-likelihood is flat in δ over roughly $[0.7, 1.0]$, so δ is only weakly identified from two periods of choices. (The BFGS routine’s convergence flag reflects this flat ridge; the gradient norm at the reported optimum is $\sim 2 \times 10^{-4}$.)

The period-1 trade-off. Schooling delivers *no* current payoff ($R_3 = \epsilon_3$) but is the only way to acquire education $G(2) = 1$, which sharply raises age-2 white-collar utility (via $\beta_1 = 0.37$) and modestly raises blue-collar utility ($\beta_2 = 0.12$). Blue-collar work pays the immediate intercept α_2 *and*

builds high-return experience ($\gamma_2 = 0.35$), making it attractive on both margins. White-collar work pays α_1 now but accumulates little useful experience ($\gamma_1 \approx 0$), so its appeal in period 1 rests mostly on the current flow. The agent therefore weighs immediate utility against the discounted option value of human capital, and how strongly that option value pulls her toward school is governed by δ .